

Report team 17

1. 10 Points about the thesis.

1. The author states his motivation for this work, object of the investigation (quadrotor Linkquad) and the goal, which is to design an autonomous control loop for a given quadrotor.
2. To achieve the stated goal the author must model the quadrotor. This includes the derivation of adequate mathematical equations that describe the object.
3. The author proceeds with a description of existing control techniques like PID, LQR, h-infinity and nonlinear control.
4. To correctly describe a quadrotor with a model it is necessary to approximate physical parameters like moments of inertia, motor dynamic, aerodynamics properties, etc.
5. One of the crucial steps toward the control loop is a nonlinear mathematical model that describes the quadrotor that will be used to evaluate effectiveness and develop control loop .
6. To have a working control loop the author evaluated limitations of the firmware (only PI control loops). It is a crucial bridging point between conceptual MATLAB/Simulink models and actual hardware of the quadrotor.
7. As next step they described a control strategy (multiple inner and outer loops for various state variables) that will allow him to implement complete position control. From the chosen strategy he designed a mixed strategy for sending commands to engines.
8. The final step of developing a control loop is linearization of the mathematical model, derivation of the state equations and testing control loop with vertical, yawing, and longitudinal movement.
9. Testing of the designed control loop was done with the attitude controllers due to various limitations of available software and hardware at the time of initial development.
10. The author concludes by stating that his work was successful but too simplified due to various reasons. Despite simplicity, the autonomous control loop was successfully implemented and tested on real-world hardware.

2. Author's algorithm

- a) Define technical goals and requirements for control loop of quadrotor.
- b) Derive the symbolic mathematical model of the quadrotor.
- c) Analyze existing control techniques.
- d) Calculate or approximate physical parameters of the quadrotor.
- e) Assess technical parameters of hardware and firmware.
- f) Linearize mathematical model to derive state equations.
- g) Design control loop that corresponds to goal and respects limitations.
- h) Test control loop in software for various modes.
- i) Test control loop with hardware.
- j) Evaluate results.

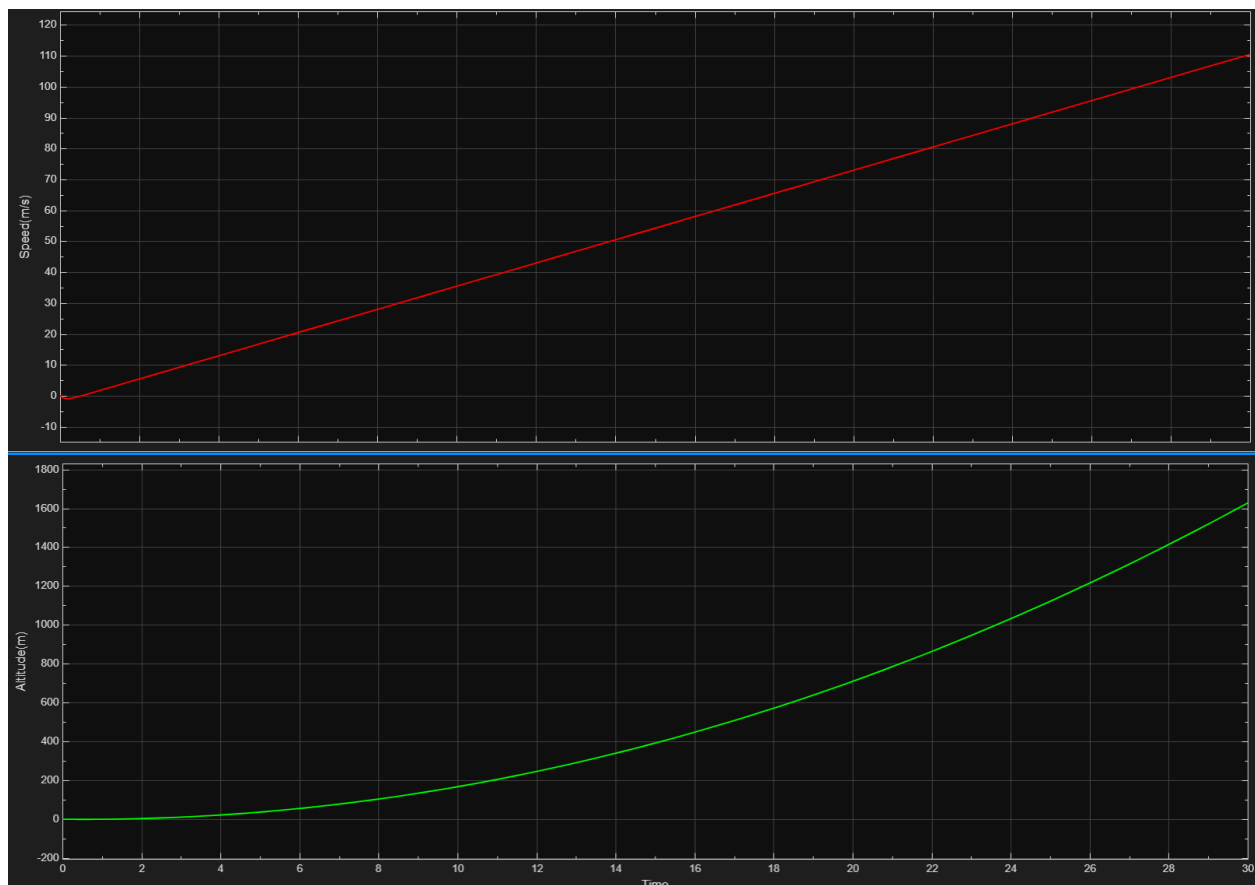
3. Summary

The student wanted to design a control loop aka. “controller” for the Linkquad quadrotor. They used well established and understood mathematical models for aerial vehicles that exhibit 6DOF (Euler's angles, linearization, state space); they clearly understood limitations of the existing hardware and firmware and from there were able to choose the correct strategy on how to design control loops. As a result they developed simplified but working control loops (PI control loops) for a given quadrotor.

4. Nonlinear model

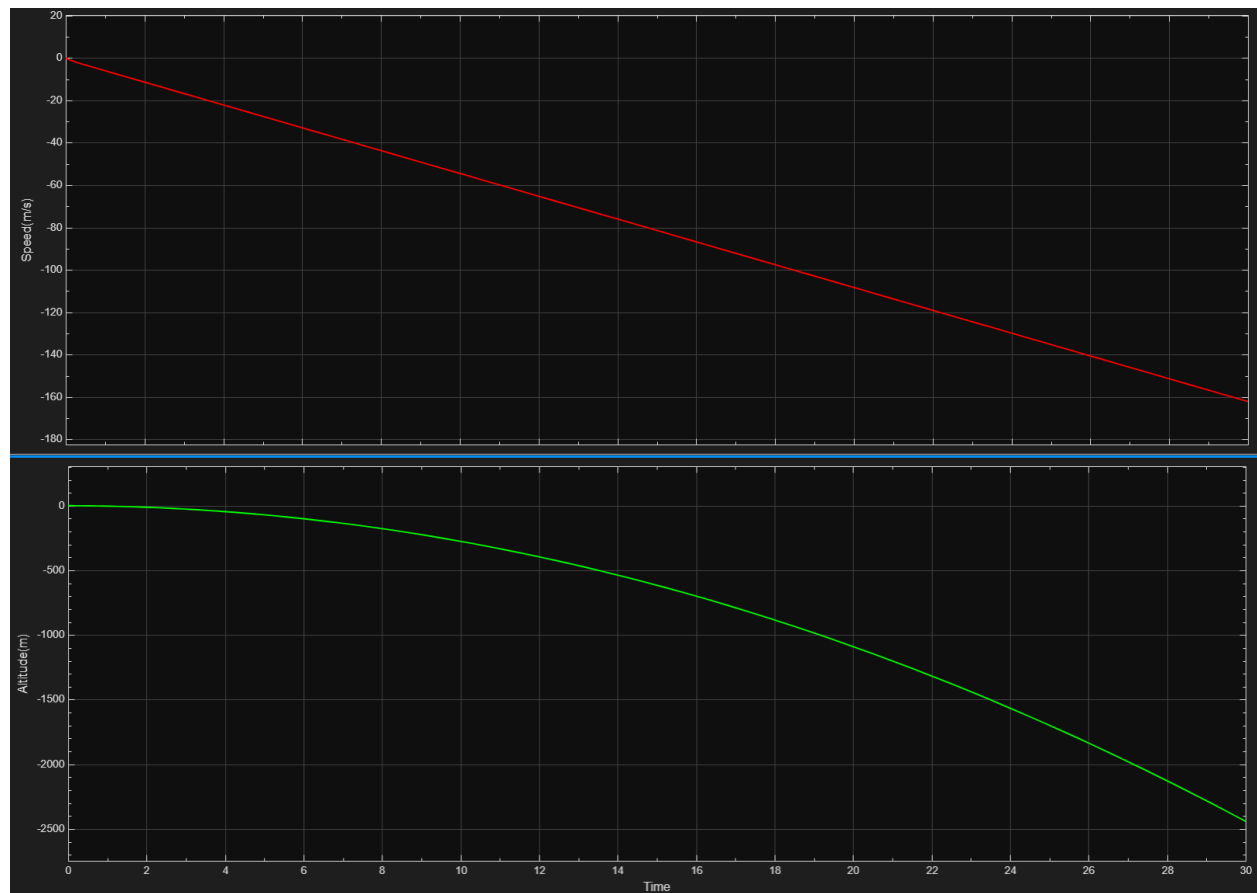
In attachment *.m files of nonlinear models from Chapter 5 of the aforementioned thesis.

To assess if the model works Step function with **Final value** of 700 was used as step value does not generate enough thrust to lift the quadrotor and overcome gravity force. Step function corresponds to angular momentum of the rotors.



Picture 1. Speed (**red**) and altitude (**green**) of the copter. Lift off.

Step function Final value is 400



Picture 2. Speed (red) and altitude (green) of the copter. As there is no ground and not enough thrust, the quadrotor is "falling".